

## FINAL EXPLANATION: BIOLOGY GRADE 10

*Teacher Key: Exemplar Answers and Marking Guide*

### FINAL EXPLANATION: BIOLOGY GRADE 10 - CHEMICALS OF LIFE: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

#### INSTRUCTIONS GIVEN TO STUDENTS

You have finished all 6 lessons of the Chemicals of Life unit. Now write your COMPLETE EXPLANATION of the two children in Kisumu. Child A is a Standard 5 pupil who eats ugali every day and has a swollen belly, thin limbs, discoloured hair and extreme tiredness (kwashiorkor). Child B lives in the same village, eats githeri regularly and is healthy and energetic. Both children eat a similar quantity of food.

DRIVING QUESTION: Why do two children in Kisumu eating the same amount of food every day end up with completely different bodies, and what does that tell us about what food is really made of?

What to use: your notebook, your Lesson 1 prediction and first model, your food test results table, your molecular models from Lesson 4, your Driving Question Board cards and your final group poster.

Write in your own words in the space under each prompt. Use specific evidence from your lessons (what you saw in the tests and what you built), and give reasons (because...) that link each idea back to the two children. Answer in full sentences. You may add labelled diagrams. Write a focused answer to each part; about 3-4 typed pages or 4-5 handwritten pages in total is enough.

You will be marked on: understanding of the chemicals of life, use of evidence from the food tests and models, explaining the phenomenon, scientific reasoning, and applying your ideas to a real Kenyan diet.

#### Part 1: What Is Food Made Of?

**Food is not just 'food'. (a) Name the six chemicals of life and give the main job of each in the body. (b) For each chemical, give one Kenyan food that is a good source. (c) Think about ugali and githeri. Using what you learned in Lessons 1 and 2, compare the main chemicals you would expect in each food and explain how this could matter for the two children.**

The six chemicals of life are: carbohydrates, which give energy (e.g. ugali, mango, sugarcane juice); proteins, which build and repair body tissues and make enzymes and hormones (e.g. beans, omena, eggs, milk); lipids, which store energy, form cell membranes and carry fat-soluble vitamins (e.g. groundnuts, avocado, coconut); nucleic acids, which carry genetic information (found in the cells of all foods, such as meat, beans and vegetables); vitamins and minerals, which are needed in small amounts to keep the body working, for example vitamin A for sight and iron for blood (e.g. sukuma wiki, milk, mango); and water, which is the solvent in which body reactions happen and transports substances. Comparing the two foods: ugali is almost entirely carbohydrate (starch) and has very little protein. Githeri combines maize (carbohydrate) with beans, which add a large amount of protein. So the two meals can be the same size but contain very different chemicals, and this could explain why the children's bodies differ.

## Part 2: Food Tests as Evidence

**Describe the food tests whose results you used in this unit. (a) For starch, protein and lipids, which test gave the evidence and what positive result is seen for each? (b) What results were recorded for ugali, for beans or githeri, and for the lipid-rich foods you tested? (c) Why is a food test better evidence than simply looking at the food?**

For starch the evidence was iodine solution: a positive result is a blue-black colour. For protein it was the Biuret test: a positive result is a purple colour. For lipids I used the translucent (grease-spot) test, where fat leaves a see-through spot on paper, and the ethanol emulsion test, where a cloudy white emulsion forms. Results: ugali turned blue-black with iodine, so it contains starch, but the Biuret test on ugali extract showed no purple colour, so no protein was detected. The Biuret test on beans extract turned purple, so beans contain protein, and the data showed githeri contains both starch and protein. In the lipid tests, ugali tested negative in both tests, while groundnuts, avocado, coconut flesh and whole milk tested strongly positive. Lipid results for githeri were recorded in my Lesson 3 table, and I read them from there. Tests are better evidence than looking because two foods can look similar but contain very different chemicals; the colour changes give observable proof of what is detected or not detected.

## Part 3: Structure and Function of Carbohydrates and Proteins

**Use your Lesson 4 models. (a) Describe how starch is built up, starting from glucose. (b) Describe how a protein is built up, starting from amino acids, and name the bond that joins them. (c) Which element is found in amino acids but not in carbohydrates? (d) Explain why starch cannot do the jobs that protein does in the body.**

Starch is a polysaccharide. Many glucose units (monosaccharides) are joined by glycosidic bonds; two joined units make a disaccharide such as sucrose (in chai), and long chains make starch (in ugali). Animals store glucose in a different polysaccharide, glycogen, in muscle and liver. A protein is made of many amino acid monomers joined by peptide bonds; in my model, bottle tops were the amino acids and straws were the peptide bonds, and two joined made a dipeptide. Amino acids contain nitrogen (N) as well as carbon, hydrogen and oxygen, whereas carbohydrates contain only C, H and O. Starch is made only of glucose and contains no nitrogen and no amino acids, so the body cannot use it to build proteins. It supplies energy, but it cannot do protein's jobs, which include building muscle and making haemoglobin, albumin and keratin.

## Part 4: Explaining the Two Children in Kisumu

**Write a Claim-Evidence-Reasoning explanation of why Child A (ugali) and Child B (githeri) have different bodies even though they eat a similar quantity of food. Your explanation must (a) make a claim, (b) use evidence from your food tests and models, and (c) explain how the chemical composition of Child A's food leads to each of these signs: swollen belly, thin limbs,**

**Claim:** The children differ because of the chemical composition of their food, not the quantity. Ugali supplies carbohydrate but almost no protein, while githeri supplies both. **Evidence:** iodine turned blue-black in ugali (starch present), but the Biuret test on ugali showed no purple colour (no protein detected). The Biuret test on beans turned purple, so beans supply protein, and githeri contains both starch and protein. My Lesson 4 models showed that proteins are made of amino acids containing nitrogen, which starch does not contain. **Reasoning:** Child A gets energy from starch but no amino acids, so the body cannot make new proteins. Without albumin, fluid is not held properly in the blood, so it collects in the belly, causing a swollen belly (oedema). Without muscle proteins, muscles waste, giving thin limbs. Without keratin, hair loses its colour and strength. Without enough protein for enzymes and other body functions, the child is extremely tired. This condition is kwashiorkor. Child B gets amino acids from the beans, so the body can build and repair tissues and stays healthy and energetic. (d) Ugali tested negative for lipids in both lipid tests, so it contains little or no fat. The lipid tests cannot show whether vitamins are present or absent. However, from Lesson 3 I learned that fat-soluble vitamins (A, D, E, K) need dietary fat to be absorbed, so a diet very low in fat could also make it harder to absorb them. This could add problems such as night blindness (vitamin A), weak bones (vitamin D) and poor immunity, alongside the protein

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| discoloured hair and tiredness. (d) Look at your Lesson 3 lipid test results for ugali. What do they show, and what can they NOT show on their own? | problem. |
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### Part 5: Applying Your Learning to a Kenyan Diet

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| <p><b>(a) Match each condition to the chemical of life it is linked to: kwashiorkor, marasmus, scurvy, rickets, anaemia. (b) A family in the Rift Valley eats only ugali for every meal. Using what you know from your tests and models, predict which chemicals of life are plentiful and which are likely to be low, and what health problems may result. (c) Suggest two Kenyan foods you would add and explain why each helps. (d) Look back at your Lesson 1 prediction: what has changed in your thinking?</b></p> | <p>(a) Kwashiorkor: protein. Marasmus: not enough total food energy, including both protein and carbohydrate. Scurvy: vitamin C. Rickets: vitamin D and calcium. Anaemia: iron. (b) A diet of only ugali is plentiful in carbohydrate (starch) and gives energy. My tests showed it has very little protein and no lipids. Other vitamins and minerals were not tested, so I cannot say from my results which are missing, but a diet of one food is unlikely to be balanced. The family could develop protein deficiency such as kwashiorkor, and low fat intake could make it harder to absorb fat-soluble vitamins. (c) Add beans (making githeri) because they supply protein with amino acids for building and repairing the body. Add sukuma wiki because it provides vitamins and minerals that help the body work and defend against disease. Groundnuts or avocado could also be added for lipids and to help absorb fat-soluble vitamins. (d) In Lesson 1 I thought the difference was something 'missing' in ugali but I could not say what. Now I can explain with evidence that ugali is starch with no detectable protein, that protein is made of amino acids containing nitrogen, and that each chemical of life has a job no other chemical can do. So food quantity is not enough; a balanced mixture of chemicals is needed, which is what the two children in Kisumu show.</p> |
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### RUBRIC

| Criterion                                  | Excellent (4)                                                                                                                                                                                                                                                                    | Proficient (3)                                                                                                            | Developing (1–2)                                                                                                      |
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| Understanding of the chemicals of life     | Correctly names all six chemicals, gives the role and a Kenyan food source for each, and accurately describes how carbohydrates and proteins are built from monomers, including the nitrogen difference.                                                                         | Names most chemicals with roles and food sources; describes carbohydrate and protein structure with minor gaps or errors. | Names few chemicals or gives vague or incorrect roles; structure of carbohydrates or proteins is missing or confused. |
| Use of evidence from food tests and models | Accurately cites specific results (iodine blue-black in ugali, no purple Biuret in ugali, purple in beans, lipid tests negative in ugali and positive in groundnuts, avocado, coconut and milk), names the correct colour for each, and says what the tests can and cannot show. | Cites most test results and colours correctly, with some missing detail.                                                  | Evidence is general, missing or incorrect.                                                                            |

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| Explaining the phenomenon            | Gives a clear Claim-Evidence-Reasoning explanation linking low protein to each sign (swollen belly, thin limbs, hair, tiredness) and explains that composition, not quantity, causes the difference.                              | Explains the difference with protein as the cause but links to only some signs or gives limited reasoning. | States that ugali is bad or lacks something without a chemical explanation, or focuses on quantity. |
| Scientific reasoning and application | Logically connects evidence to conclusions, separates what the tests showed from what is inferred about lipids and vitamins, applies ideas to a Kenyan diet with sensible food suggestions, and reflects on how thinking changed. | Applies ideas to a diet with reasonable suggestions; reasoning mostly clear.                               | Applications are unclear or unsupported; little reasoning or reflection.                            |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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